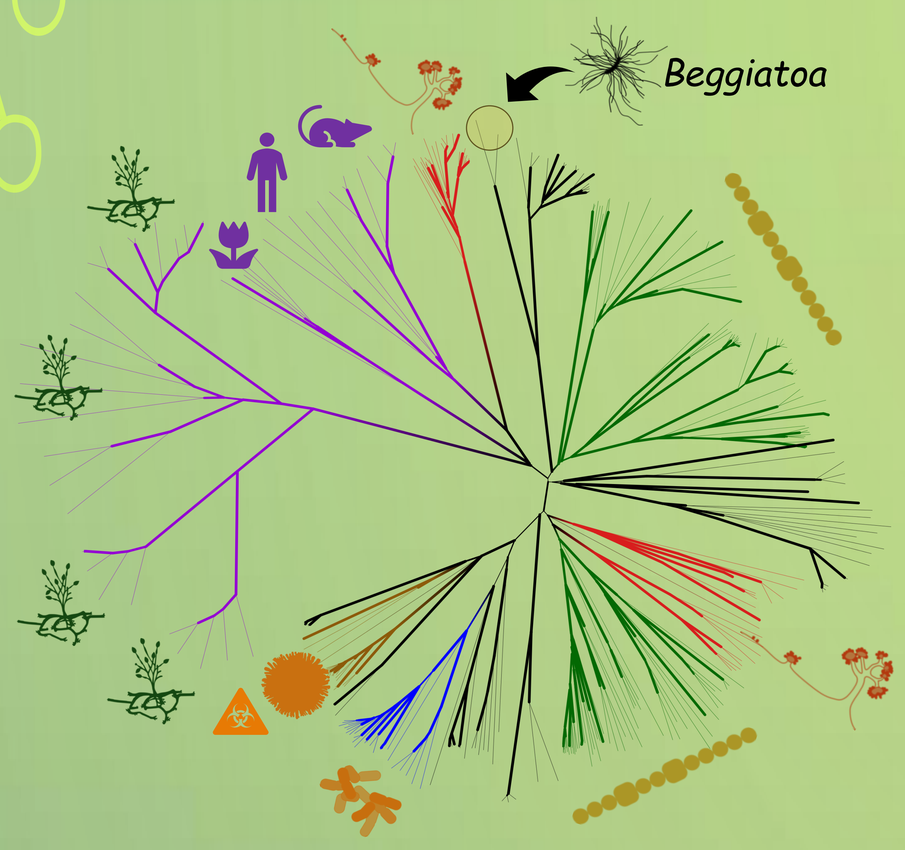
Lipoxygenase sequences in *Beggiatoa spp.* suggest a new life strategy in giant sulfur bacteria

Georgy Kurakin

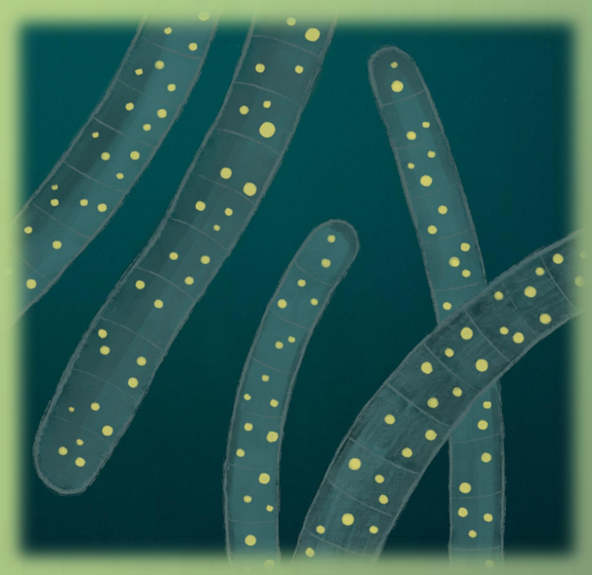
Pirogov Russian National Research Medical University, Moscow, Russia, **email:** georgykurakin@gmail.com

In our previous paper, we had found that lipoxygenase is associated even with such simple types of multicellularity across both bacteria and eukaryotes. If our assumptions are true, **lipoxygenase could be the most universal biochemical marker of multicellularity we know**.



The statistical and phylogenetic picture observed was created mainly by **cyanobacteria**, **myxobacteria**, **dictyostelid slime moulds** and **complex multicellular eukaryotes**.
But there was one more multicellular bacterial taxon in our sample — ***Beggiatoa***, a genus of giant sulfur bacteria.
Their contribution to the statistical picture was **so negligible that we omitted them that time.**

Beggiatoa spp.



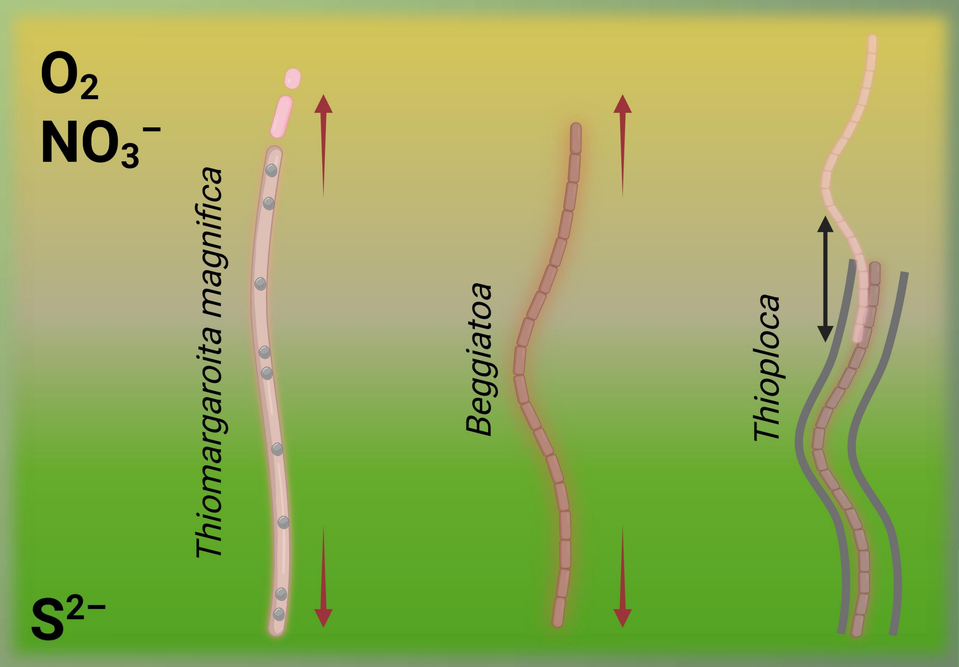
Discovery of *Thiomargarita magnifica* — the biggest-ever bacterium — urged us to revisit our recent bioinformatic finding of lipoxygenase enzyme in *Beggiatoa*, the **closest relatives** of *Thiomargarita*.

Thioploca relies more on its **motility**, *Thiomargarita magnifica* relies on its **cell complexity**, and only *Beggiatoa* has the only strategy to bridge the substrate pools — its **multicellularity**. This could explain why it needs LOXs in contrast to other *Thiotrichales* and breaks the statistical pattern.

Image credit:
*Anastasiia Samoukina (**inst:** @samoukanastasia) and Matt Crook/**PhyloPic**/ CC BY-SA 3.0*

Created with BioRender.com

Tree visualization in iTOL Interactive Tree Of Life



Lipoxygenases in *Beggiatoa* could be linked to their multicellularity.

Read more:
Biochemistry (Moscow) (2023) 88: 842-845
https://doi.org/m57v

Read more:
Springer Nature Research Communities blog post
https://bit.ly/4bud8Fc